

DUSHYANT JOSHI

AI/ML Engineer | Generative AI | MLOps | ML Infrastructure | LLM Evaluation

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SUMMARY

AI/ML Engineer with 2 years building, deploying and operating production **ML and LLM systems on AWS**. Owns the infrastructure side of the ML lifecycle: **containerized** async Python services shipped through **CI/CD**, end-to-end **monitoring and distributed tracing** with **OpenTelemetry**, automated **evaluation gates** that block bad model changes before release, and **cost optimization** that cut compute waste on production workloads. Owns systems from **data pipeline** through **deployment, monitoring** and production **troubleshooting**.

SKILLS

Generative AI and Agents: RAG, Agentic and Multi-Agent Systems, ReAct, Function and Tool Calling, Prompt Engineering, Fine-Tuning, Guardrails, Knowledge Graphs, MCP

LLM Frameworks and Models: LangChain, LangGraph, OpenAI API, Anthropic Claude API, GPT-4o, Llama-3.2, LiteLLM, Pinecone, Qdrant, FAISS, Hybrid Retrieval, Reranking

LLM Evaluation: Offline and Online Evals, LLM-as-a-Judge, G-Eval, RAG Triad, Faithfulness, Recall@k, NDCG, Hallucination Testing, Regression and Safety Evals

ML Lifecycle and Deployment: Model Deployment, Model Routing, Prompt and Config Versioning, Evaluation Gating in CI, Deployment Troubleshooting

Monitoring and Observability: OpenTelemetry, Distributed Tracing, Langfuse, P95 Latency Monitoring, Token and Cost Telemetry, Error Tracking, Alerting

Data Engineering: SQL, Fact Table Modelling, Async Ingestion Pipelines, Data Validation and Quality, PostgreSQL, MySQL, MongoDB

Programming and Backend: Python, Async Python, FastAPI, Django, REST Microservices, OOP, Data Structures and Algorithms

Cloud and Infrastructure: AWS, Docker, Containerization, CI/CD Pipelines, Git, Linux, Shell

Currently Learning: Kubernetes, Terraform, Apache Spark, Prometheus, Grafana (Q4 2026)

EXPERIENCE

Analyttica Datalab, AI/ML Engineer

Mar 2025 - Present | Bengaluru, India

- Deployed and operated the production AI services powering **Copilot**, an enterprise AI assistant: async **REST microservices** in **FastAPI**, **containerized with Docker** and released to **AWS** through **CI/CD**, serving **1,500+ weekly production queries**.
- Owned **monitoring and observability**, instrumenting **OpenTelemetry** and Langfuse **distributed tracing** across a five-stage pipeline (orchestrator, grounding, SQL generation, validation, response) to capture **latency, error, token and cost metrics per stage**, cutting production **troubleshooting time 40%**.
- Built an **AI usage and cost-analytics data pipeline** end to end, from Langfuse ingestion into **MySQL fact tables** to layered **FastAPI** APIs; the analysis showed **20% of workloads consumed 60% of compute**, driving a **model-routing and cost-optimization** strategy.
- Architected production **RAG** pipelines and an API integration router with **LangChain**, **GPT-4o** and **Pinecone**, driving a **70% increase in customer engagement** at **sub-200ms retrieval latency** across a **10,000+ document corpus**.
- Designed an enterprise-scale **KPI Knowledge Graph** unifying **heterogeneous SQL, API and unstructured sources** with validation rules, automating recurring **data processing workflows** for business teams.
- Worked **cross-functionally** with business and analytics stakeholders to deliver **5+ enterprise POCs** from requirements through deployed, monitored systems.

American Chase, Software Developer

Nov 2024 - Mar 2025 | Indore, India

- Delivered **Python** backend services for an in-house payroll and timesheet platform covering employee, attendance and payroll-processing workflows, working in an Agile team across debugging, testing, **Git**-based development and deployment.

PROJECTS

Pro-RAG, Production ML System

Python, FastAPI, Qdrant, PostgreSQL, Docker

- Engineered a **scalable retrieval system architected for 10M documents at 50 QPS**, with an **async ingestion pipeline** (parse, chunk, embed, index), **semantic caching**, **multi-tenant quotas** and **PostgreSQL cost metering**.
- Built **automated evaluation gating** into the release path: Recall@k and NDCG for retrieval, faithfulness and answer relevance for generation, with **regression checks that block configuration changes** that degrade quality; added prompt-injection detection, **PII guardrails** and confidence-based refusal.

Multi-Modal Intelligent Query Agent

Python, LangGraph, FastAPI, PostgreSQL

- Built an autonomous agent with **LangGraph** using tool calling, planning and memory over SQL and unstructured data; the **async backend cut query-resolution time 60%** versus the synchronous baseline.

CERTIFICATIONS AND ACHIEVEMENTS

- Anthropic Academy (Claude)**, 5 course badges, Sept 2026: Claude 101, Claude Platform 101, Claude Code 101, Claude Code in Action, Introduction to Claude Cowork. Preparing for Claude certification. **HackerRank SQL Certified**.
- Self-directed study in **LLM and RAG evaluation**: offline and online evals, LLM-as-a-Judge, G-Eval, RAG Triad, retriever and generator testing, safety and regression evals.
- Solved **350+ LeetCode and GeeksforGeeks problems** (algorithms, **SQL**). National-level Volleyball player.

EDUCATION

Bangalore Institute of Technology, B.E. Electronics and Instrumentation | 2020 - 2024 | CGPA: 7.0/10